

Colby Dobson

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EDUCATION

University of San Francisco, San Francisco, CA

- President, Club Men's Volleyball August 2023 - May 2026
- B.S. in Computer Science August 2021 - May 2025
 - *Minor in Mathematics*
- M.S. in Computer Science; AI Concentration August 2023 - May 2026

COURSEWORK

Operating Systems, Systems Foundations, Advanced Algorithms, Compilers, Software Development Lifecycle, Cloud Computing, Programming Language Paradigms, Software Development, DevOps, Bayesian Deep Learning, Machine Learning, Programming Synthesis, Advanced Linear Algebra, Linear Regression, Full Stack Web-Dev

PROJECTS

Embedded Systems Arcade Game Spring 2025 - Present

- Architecting and developing a distributed embedded arcade system across Raspberry Pis with custom hardware interfaces, real-time input arbitration, and synchronized communication
- Implemented fault-tolerant low-level communication, GPIO/UART signaling, and state-coordination protocols for reliable timing-sensitive multi-device interaction under I/O and resource constraints

RISC-V Emulator Spring 2023

- Emulated a RISC-V processor with C to execute binary instructions/programs.
- Designed digital circuits to build a processor using computer architecture and hardware principles.

Reverse Proxy Fall 2024

- Led a team of engineers in partnership with Consilient Labs to develop a secure, scalable, and efficient reverse proxy as the primary layer of security for a production API
- Coordinated with the client to align system design with their needs, prioritizing security through fine-grained permission controls and robust access management

Alpha-Zero Style Go Engine Spring 2026

- Built a Go engine from scratch, progressing to an AlphaZero-style self-play system with policy/value CNNs, MCTS, champion-gated training, and live model evaluation tooling

Agentic Assistant Spring 2026

- Building a microservice-based, distributed, multi-user AI assistant with agent-based task execution
- Designed and implemented a personalized agent memory system using vector databases, embeddings models, and RAG to persist user preferences, retrieve relevant long-term context, and ground agent responses and actions in individual user history

TECHNICAL SKILLS

Programming Languages: Rust, C++, C, Python, Java, TypeScript, JavaScript, Assembly (RISC-V)

Systems: Linux, Embedded Systems, UART, TDMA, GPIO, Concurrency, Multithreading, Hardware/Software Integration, Computer Architecture, Digital Logic, Protocol Design, Distributed Systems, Networking

Infrastructure: Git, Docker, AWS, CI/CD, Microservices, Actions, K8s, REST, SQL, noSQL, OAuth, Prometheus + Grafana

AI/ML: CNN, Transformer, PyTorch, Regression, LLM Integration, Agent Workflows & Infrastructure, RAG, Tools, Skills

WORK EXPERIENCE

Research Assistant June 2026 - Present

- Developed and evaluated a foundation model for robust redshift prediction across astronomical surveys
- Designed controlled experiments, cross-survey benchmarks, and scalable HPC training pipelines

University of San Francisco, Computer Science Tutor & TA August 2024 - December 2025

- Quickly debug and identify issues while explaining complex topics at an elementary level

Sláinte Volleyball Club, Head Coach September 2021 - July 2025

- Coordinated practices and tournaments, fostered development, and maintained a good environment